Every rank in the alternating group interval

Human Compression project: principal recovery of L038

Author integration; distinct review pending

Theorem 1. *For every integer $n \geq 12$ and every integer $3 \leq r \leq \lfloor (n-1)/2 \rfloor$, the alternating group A_n has a string C-group representation of rank r.*

This is Theorem 1.1 of Fernandes and Leemans, *String C-group representations of alternating groups*, in the fixed edition arXiv:1810.12450v2, 4 January 2019. The two companion proofs establish the required permutation constructions. Here we assemble their conclusions with the already acquired contraction theorem and two existing degree-twelve witnesses. The maximality of the interval is a separate earlier theorem; it is not asserted as a new conclusion of this recovery.

1 The precise acquired interfaces

A string C-marking $(s_0, \ldots, s_{R-1})$ is irreducible if every adjacent product has order greater than two. Put $p_i = |s_{i-1}s_i|$. The acquired proof of Brooksbank–Leemans rank contraction gives the following statement: if $p_3, \ldots, p_{t+2}$ are odd, where $0 \leq t \leq R-3$, the same group has C-markings of all ranks $R-t, \ldots, R$. The ordering may first be reversed. Every contraction preserves irreducibility. We use this proved indexing, rather than the extraneous p_2 test in the source's printed iteration formula.

Two earlier maximal-rank constructions are retained as inputs. They are exactly the inputs already recorded in the acquired L037 application *Rank intervals from one string marking*, Section 2, and are explicitly attributed to the earlier literature in the fixed L038 source:

- For $n \equiv 1 \pmod 4$, $n \geq 13$, rank $R = (n-1)/2$, with type $[10, 3^{R-2}]$. The source recalls the prior alternating construction in Theorem 3.2 and uses it in Theorem 4.6.
- For even $n \geq 14$, rank $R = (n-2)/2$, with type $[5, 6, 3^{R-6}, 6, 6, 3]$. This is the earlier maximal-rank construction explicitly recalled at the start of the proof of source Theorem 4.1.

These include generation and the full intersection property; they are not merely assertions about adjacent orders. Contraction therefore supplies every rank $3, \ldots, R$ in the first case and every rank $6, \ldots, R$ in the second. The latter uses exactly $R-6$ consecutive odd bonds beginning at p_3; when $R = 6$, no contraction is required.

Our reviewed companion *Finite end groups for alternating group constructions* proves, with faithful finite end certificates, the following four families from the source:

Degree	Rank	Type	Source theorem
$n \equiv 2\ (4),\ n \geq 10$	5	$[5, 5, 6, n-5]$	4.3
$n \equiv 0\ (4),\ n \geq 12$	5	$[3, 4, 6, n-7]$	4.5
$n \equiv 0\ (4),\ n \geq 16$	4	$[3, 12, \mathrm{lcm}(n-8, 6)]$	4.4
$n \equiv 3\ (4),\ n \geq 15$	6	$[n-10, 6, 3, 5, 3]$	4.13

In the first row reverse the marking: its tested bonds are five and five, giving ranks four and three. In the third row reversal gives third bond three and hence rank three. In the fourth row the tested bonds are three, five and three, giving ranks five, four and three. All adjacent orders are greater than two, so the irreducibility hypothesis is present in each application.

The second companion, *Signed permutation groups at the ends of alternating group constructions*, proves the exact source Theorems 4.8–4.10 by two uniform signed-group calculations and explicit parity intersections. Its conclusions are:

- If $r \geq 7$ is odd and $k \geq 1$, $A_{2r+1+4k}$ has a rank-r C-marking. The case $k = 0$ is the earlier maximal-rank seed, explicitly retained through source Theorem 4.7.
- If $r \geq 8$ is even and $k \geq 0$, $A_{2r+3+4k}$ has a rank-r C-marking, including the $k = 0$ boundary.

The companion proves their exact displayed types as well. It uses the earlier odd maximal-rank seed only where it explicitly appears, and does not assume the principal theorem being recovered.

2 Degree twelve from the existing catalogue

Two tuples from the already accepted L071 alternating-group census are enough for the remaining degree-twelve boundary. They are listed here so the existence assertion is independently readable:

$$\begin{aligned} a_0 &= (5\ 6)(7\ 8)(9\ 10)(11\ 12), \\ a_1 &= (3\ 5)(4\ 6)(8\ 9)(10\ 11), \\ a_2 &= (1\ 3)(2\ 4)(5\ 7)(6\ 8), \end{aligned}$$

and

$$\begin{aligned} b_0 &= (9\ 10)(11\ 12), \\ b_1 &= (5\ 6)(7\ 9)(8\ 11)(10\ 12), \\ b_2 &= (1\ 5)(2\ 6)(3\ 7)(4\ 8)(9\ 10)(11\ 12), \\ b_3 &= (2\ 3)(7\ 8)(9\ 11)(10\ 12). \end{aligned}$$

The recorded ranks are three and four, with types $[12, 20]$ and $[6, 8, 6]$. Every mark is even, each tuple generates a group of order $12!/2$, and all $8^2 = 64$ and $16^2 = 256$ ordered pairs of subset subgroups satisfy the intersection property. Their string commutations also hold. Thus these are representations of A_{12}. The accompanying small certificate checks these two already recorded objects, without performing a new classification. In the retained L071 records they are rank-three row 43, involution indices $(117, 1903, 23898)$, and rank-four row 20, indices $(1, 122, 32811, 5368)$. The exact predecessor record file and its accepted review are bound in the dependency manifest. Rank five at $n = 12$ is already the second low-rank family above.

3 Completion of the interval

Proof of Theorem 1. For $n = 12$, the two recorded witnesses and the rank-five family give all three required ranks. If $n \equiv 1 \pmod 4$, the first maximal-rank seed and contraction give the whole interval.

Suppose next that $n \geq 14$ is even. The second maximal-rank seed and contraction give all ranks at least six through $R = (n - 2)/2$. When $n \equiv 2 \pmod 4$, the first low-rank family and

its two contractions give ranks three, four and five. When $n \equiv 0 \pmod 4$, necessarily $n \geq 16$; the second and third low-rank families give rank five and ranks four and three, respectively. These intervals meet and cover every $3 \leq r \leq R$.

Finally let $n \equiv 3 \pmod 4$, so $n \geq 15$ and $R = (n-1)/2$ is odd. The rank-six family and its contractions give ranks three through six. For every odd $7 \leq r \leq R$, put $k = (R-r)/2$. This is a nonnegative integer and $n = 2r + 1 + 4k$; use the higher odd family if $k \geq 1$ and the retained maximal-rank seed if $k = 0$. For every even $8 \leq r \leq R-1$, put $k = (R-1-r)/2$. Again k is a nonnegative integer, and $n = 2r+3+4k$; the higher even family applies. If $R = 7$, the even list is empty, as required. Together these give every rank from three through R. □

4 What is recovered and what is retained

The whole principal existence statement, source Theorem 1.1, is recovered. The constructions of source Theorems 4.3–4.5, 4.8–4.10 and 4.13 are recovered with their exact types by the two companions. For source Theorems 4.1 and 4.6, this integration gives their rank-existence conclusions from earlier maximal seeds. For 4.2, 4.11, 4.12 and 4.14, it likewise gives the required rank existence, using contraction instead of the source's different lower-rank diagrams. Their exact types and every auxiliary proposition are not claimed as additional recoveries.

The earlier maximal constructions, standard end criterion and polytope correspondence retain their original attribution. The rank-contraction theorem is the acquired proof of Brooksbank–Leemans, not a new theorem in this integration. Our work lies in the complete end calculations, the faithful finite reductions with their constraints, and the assembly which eliminates separate lower-rank constructions. This is not an open-conjecture solution or a historical priority claim. No new Lean formalization or complexity measurement is part of this intake recovery.

References

[1] M. E. Fernandes and D. Leemans, *String C-group representations of alternating groups*, arXiv:1810.12450v2, 4 January 2019; Ars Math. Contemp. 17 (2019), 291–310.

[2] P. A. Brooksbank and D. Leemans, *Rank reduction of string C-group representations*, arXiv:1812.01055v2, 3 May 2019; Proc. Amer. Math. Soc. 147 (2019), 5421–5426.

[3] D. Leemans and M. Mixer, *Algorithms for classifying regular polytopes with a fixed automorphism group*, Contributions to Discrete Mathematics 7(2) (2012), 105–118. The acquired L071 census supplies the two literal witnesses above.